

Q1. A particle moves in $x - y$ plane under the influence of a force $\vec{F}$ such that its linear momentum is

$\vec{p}(t) = \hat{i} \cos(kt) - \hat{j} \sin(kt)$. If k is constant, the angle between $\vec{F}$ and $\vec{p}$ will be :

- (1) $\frac{\pi}{4}$ (2) $\frac{\pi}{6}$
(3) $\frac{\pi}{2}$ (4) $\frac{\pi}{3}$

Q2. What is the dimensional formula of ab^{-1} in the equation $\left(P + \frac{a}{V^2}\right)(V - b) = RT$, where letters have their usual meaning.

- (1) $[M^{-1} L^5 T^3]$ (2) $[M^6 L^7 T^4]$
(3) $[ML^2 T^{-2}]$ (4) $[M^0 L^3 T^{-2}]$

Q3. A man carrying a monkey on his shoulder does cycling smoothly on a circular track of radius 9 m and completes 120 revolutions in 3 minutes. The magnitude of centripetal acceleration of monkey is (in m/s^2) :

- (1) $57600\pi^2 ms^{-2}$ (2) Zero
(3) $4\pi^2 ms^{-2}$ (4) $16\pi^2 ms^{-2}$

Q4. A heavy box of mass 50 kg is moving on a horizontal surface. If co-efficient of kinetic friction between the box and horizontal surface is 0.3 then force of kinetic friction is :

- (1) 1.47 N (2) 147 N
(3) 14.7 N (4) 1470 N

Q5. A body is moving unidirectionally under the influence of a constant power source. Its displacement in time t is proportional to :

- (1) t (2) $t^{3/2}$
(3) t^2 (4) $t^{2/3}$

Q6. A satellite revolving around a planet in stationary orbit has time period 6 hours. The mass of planet is one-fourth the mass of earth. The radius orbit of planet is : (Given = Radius of geo-stationary orbit for earth is

$4.2 \times 10^4 \text{ km}$)

- (1) $1.4 \times 10^4 \text{ km}$ (2) $1.05 \times 10^4 \text{ km}$
(3) $8.4 \times 10^4 \text{ km}$ (4) $1.68 \times 10^5 \text{ km}$

Q7. Match List-I with List-II :

	List-I		List-II
(A)	A force that restores an elastic body of unit area to its original state	(I)	Bulk modulus
(B)	Two equal and opposite forces parallel to opposite faces	(II)	Young's modulus
(C)	Forces perpendicular everywhere to the surface per unit area same everywhere	(III)	Stress
(D)	Two equal and opposite forces perpendicular to opposite faces	(IV)	Shear modulus

Choose the correct answer from the options given below :

- (1) (A)-(IV), (B)-(II), (C)-(III), (D)-(I) (2) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
(3) (A)-(II), (B)-(IV), (C)-(I), (D)-(III) (4) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Q8. During an adiabatic process, if the pressure of a gas is found to be proportional to the cube of its absolute temperature, then the ratio of $\frac{C_P}{C_V}$ for the gas is :

(1) $\frac{5}{3}$
 (3) $\frac{3}{2}$

(2) $\frac{9}{7}$
 (4) $\frac{7}{5}$

Q9. If n is the number density and d is the diameter of the molecule, then the average distance covered by a molecule between two successive collisions (i.e. mean free path) is represented by :

(1) $\sqrt{2}n\pi d^2$
 (3) $\frac{1}{\sqrt{2}n\pi d^2}$

(2) $\frac{1}{\sqrt{2}n\pi d^2}$
 (4) $\frac{1}{\sqrt{2}n^2\pi^2 d^2}$

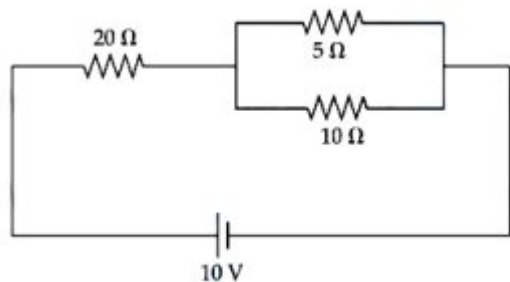
Q10. The vehicles carrying inflammable fluids usually have metallic chains touching the ground :

- (1) To protect tyres from catching dirt from ground (2) To alert other vehicles
 (3) It is a custom (4) To conduct excess charge due to air friction to ground and prevent sparking

Q11. A galvanometer of resistance 100Ω when connected in series with 400Ω measures a voltage of upto 10 V. The value of resistance required to convert the galvanometer into ammeter to read upto 10 A is $x \times 10^{-2}\Omega$. The value of x is :

- (1) 2 (2) 800
 (3) 20 (4) 200

Q12. The ratio of heat dissipated per second through the resistance 5Ω and 10Ω in the circuit given below is :



- (1) 1 : 2 (2) 2 : 1
 (3) 4 : 1 (4) 1 : 1

Q13. The electrostatic force $\left(\vec{F}_1\right)$ and magnetic force $\left(\vec{F}_2\right)$ acting on a charge q moving with velocity v can be written :

- (1) $\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{V} \times \vec{B})$ (2) $\vec{F}_1 = q\vec{B}, \vec{F}_2 = q(\vec{B} \times \vec{V})$
 (3) $\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{B} \times \vec{V})$ (4) $\vec{F}_1 = q\vec{V} \cdot \vec{E}, \vec{F}_2 = q(\vec{B} \cdot \vec{V})$

Q14. A series LCR circuit is subjected to an ac signal of 200 V, 50 Hz. If the voltage across the inductor ($L = 10\text{mH}$) is 31.4 V, then the current in this circuit is _____ .

- (1) 68 A (2) 63 A
 (3) 10 A (4) 10 mA

Q15.

Match List-I with List-II :

	List-I		List-II
	EM-Wave		Wavelength Range
(A)	Infra-red	(I)	$< 10^{-3}$ nm
(B)	Ultraviolet	(II)	400 nm to 1 nm
(C)	X-rays	(III)	1 mm to 700 nm
(D)	Gamma rays	(IV)	1 nm to 10^{-3} nm

(1) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(2) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(4) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

Q16. Given below are two statements : Statement I : When the white light passed through a prism, the red light bends lesser than yellow and violet. Statement II : The refractive indices are different for different wavelengths in dispersive medium. In the light of the above statements, chose the correct answer from the options given below:

(1) Statement I is false but Statement II is true

(2) Statement I is true but Statement II is false

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

Q17. Which of the following statement is not true about stopping potential (V_0) ?

(1) It is $1/e$ times the maximum kinetic energy of electrons emitted.

(2) It increases with increase in intensity of the incident light.

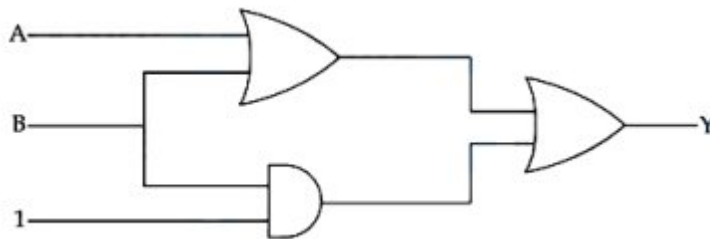
(3) It depends on the nature of emitter material.

(4) It depends upon frequency of the incident light.

Q18. The angular momentum of an electron in a hydrogen atom is proportional to : (Where r is the radius of orbit of electron)

(1) r (2) $\sqrt{r}$ (3) $\frac{1}{\sqrt{r}}$ (4) $\frac{1}{r}$

Q19. The output (Y) of logic circuit given below is 0 only when :

(1) $A = 1, B = 0$ (2) $A = 0, B = 1$ (3) $A = 0, B = 0$ (4) $A = 1, B = 1$

Q20. A vernier callipers has 20 divisions on the vernier scale, which coincides with 19^{th} division on the main scale. The least count of the instrument is 0.1 mm. One main scale division is equal to _____ mm.

(1) 0.5

(2) 2

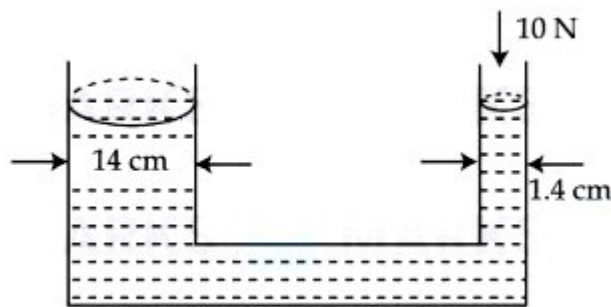
(3) 5

(4) 1

Q21. The maximum height reached by a projectile is 64 m. If the initial velocity is halved, the new maximum height of the projectile is _____ m.

Q22. A hollow sphere is rolling on a plane surface about its axis of symmetry. The ratio of rotational kinetic energy to its total kinetic energy is $\frac{x}{5}$. The value of x is _____.

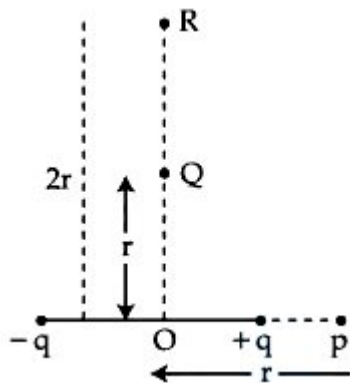
Q23.



A hydraulic press containing water has two arms with diameters as mentioned in the figure. A force of 10 N is applied on the surface of water in the thinner arm. The force required to be applied on the surface of water in the thicker arm to maintain equilibrium of water is _____ N.

Q24. A sonometer wire of resonating length 90 cm has a fundamental frequency of 400 Hz when kept under some tension. The resonating length of the wire with fundamental frequency of 600 Hz under same tension _____ cm.

Q25. The electric field at point p due to an electric dipole is E . The electric field at point R on equatorial line will be



$\frac{E}{x}$. The value of x :

Q26. A wire of resistance 20Ω is divided into 10 equal parts, resulting pairs. A combination of two parts are connected in parallel and so on. Now resulting pairs of parallel combination are connected in series. The equivalent resistance of final combination is _____ Ω .

Q27. A solenoid of length 0.5 m has a radius of 1 cm and is made up of 'm' number of turns. It carries a current of 5 A. If the magnitude of the magnetic field inside the solenoid is 6.28×10^{-3} T then the value of m is _____.

Q28. The current in an inductor is given by $I = (3t + 8)$ where t is in second. The magnitude of induced emf produced in the inductor is 12mV. The self-inductance of the inductor _____ mH.

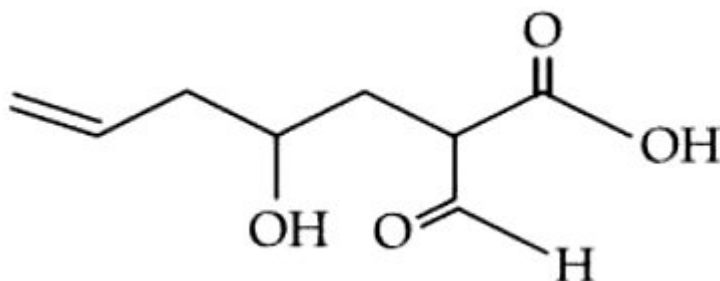
Q29. In a single slit experiment, a parallel beam of green light of wavelength 550 nm passes through a slit of width 0.20 mm. The transmitted light is collected on a screen 100 cm away. The distance of first order minima from the central maximum will be $x \times 10^{-5}$ m. The value of x is :

- Q30.** The shortest wavelength of the spectral lines in the Lyman series of hydrogen spectrum is $915\text{\AA}$. . . The longest wavelength of spectral lines in the Balmer series will be _____ $\text{\AA}$. . .
- Q31.** The number of moles of methane required to produce $11\text{ gCO}_2(\text{g})$ after complete combustion is : (Given molar mass of methane in gmol^{-1} : 16)
- (1) 0.35 (2) 0.5
(3) 0.75 (4) 0.25
- Q32.** Given below are two statements : Statement I : The metallic radius of Na is $1.86\text{ \AA}$ and the ionic radius of Na^+ is lesser than $1.86\text{ \AA}$. Statement II : Ions are always smaller in size than the corresponding elements. In the light of the above statements, choose the correct answer from the options given below :
- (1) Both Statement I and Statement II are false (2) Statement I is incorrect but Statement II is true
(3) Both Statement I and Statement II are true (4) Statement I is correct but Statement II is false
- Q33.**
- | | | | |
|---------------------------|--------------------|------------------------------|------------------------------------|
| | List - I | List - II | |
| | (A) ICl | (I) T - shape | |
| Match List I with List II | (B) ICl_3 | (II) pyramidal | Choose the correct answer from the |
| | (C) ClF_5 | (III) Pentagonal bipyramidal | |
| | (D) IF_7 | (IV) Linear | |
- options given below :
- (1) (A)-(IV), (B)-(I), (C)-(II), (D)-(III) (2) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)
(3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I) (4) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
- Q34.** Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : NH_3 and NF_3 molecule have pyramidal shape with a lone pair of electrons on nitrogen atom. The resultant dipole moment of NH_3 is greater than that of NF_3 . Reason (R) : In NH_3 , the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the N – H bonds. F is the most electronegative element. In the light of the above statements, choose the correct answer from the options given below :
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A) (2) (A) is false but (R) is true
(3) Both (A) and (R) are true but (R) is NOT the correct explanation of (A) (4) (A) is true but (R) is false
- Q35.** Given below are two statements : Statement I : On passing $\text{HCl}_{(\text{g})}$ through a saturated solution of BaCl_2 at room temperature white turbidity appears. Statement II : When HCl gas is passed through a saturated solution of NaCl , sodium chloride is precipitated due to common ion effect. In the light of the above statements, choose the most appropriate answer from the options given below :
- (1) Both Statement I and Statement II are correct (2) Statement I is correct but Statement II is incorrect
(3) Both Statement I and Statement II are incorrect (4) Statement I is incorrect but Statement II is correct
- Q36.** The correct statements from the following are : (A) The decreasing order of atomic radii of group 13 elements is $\text{Tl} > \text{In} > \text{Ga} > \text{Al} > \text{B}$. (B) Down the group 13 electronegativity decreases from top to bottom. (C) Al dissolves in dil. HCl and liberates H_2 but conc. HNO_3 renders Al passive by forming a protective oxide layer

on the surface. (D) All elements of group 13 exhibits highly stable +1 oxidation state. (E) Hybridisation of Al in $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ ion is $\text{sp}^3 \text{d}^2$. Choose the correct answer from the options given below :

- (1) (A), (C) and (E) only (2) (A) and (C) only
(3) (C) and (E) only (4) (A), (B), (C) and (E) only

Q37. The correct nomenclature for the following compound is :



- (1) 2-formyl-4-hydroxyhept-7-enoic acid (2) 2-formyl-4-hydroxyhept-6-enoic acid
(3) 2-carboxy-4-hydroxyhept-7-enal (4) 2-carboxy-4-hydroxyhept-6-enal

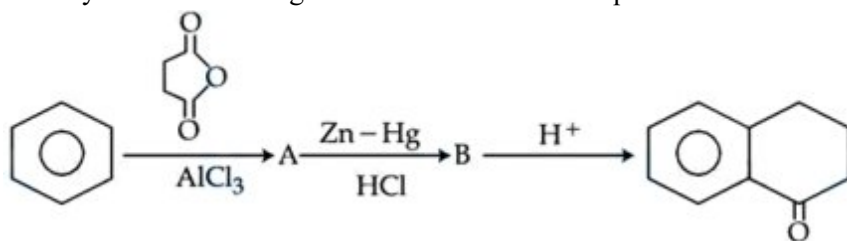
Q38.

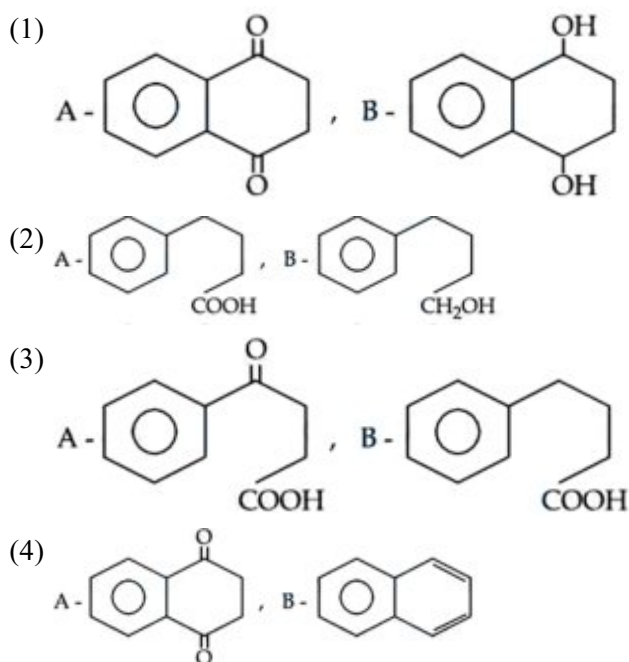
	List - I	List - II	
	(Pair of compounds)	(Isomerism)	
Match List I with List II	(A) n-propanol and Isopropanol	(I) Metamerism	Choose the
	(B) Methoxypropane and ethoxyethane	(II) Chain Isomerism	
	(C) Propanone and propanal	(III) Position Isomerism	
	(D) Neopentane and Isopentane	(IV) Functional Isomerism	

correct answer from the options given below :

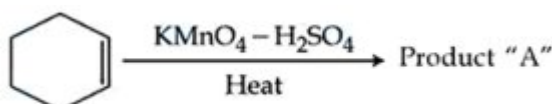
(1) (A)-(II), (B)-(I), (C)-(IV), (D)-(III) (2) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
(3) (A)-(I), (B)-(III), (C)-(IV), (D)-(II) (4) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Q39. Identify *A* and *B* in the given chemical reaction sequence :





Q40.



Consider the given chemical reaction :

Product " A " is :

- (1) picric acid (2) acetic acid
(3) adipic acid (4) oxalic acid

Q41. The quantity of silver deposited when one coulomb charge is passed through AgNO_3 solution :

- (1) 1 g of silver (2) 1 electrochemical equivalent of silver
(3) 1 chemical equivalent of silver (4) 0.1 g atom of silver

Q42. For the electro chemical cell If $E_{(\text{M}^{2+}/\text{M})}^0 = 0.46 \text{ V}$ and $E_{(\text{x}/\text{X}^{2-})}^0 = 0.34 \text{ V}$. Which of the following is correct?

- (1) $\text{M} + \text{X} \rightarrow \text{M}^{2+} + \text{X}^{2-}$ is a spontaneous reaction (2) $E_{\text{cell}} = 0.80 \text{ V}$
(3) $E_{\text{cell}} = -0.80 \text{ V}$ (4) $\text{M}^{2+} + \text{X}^{2-} \rightarrow \text{M} + \text{X}$ is a spontaneous reaction

Q43. The number of ions from the following that have the ability to liberate hydrogen from a dilute acid is ____.

 Ti^{2+} , Cr^{2+} and V^{2+}

- (1) 2 (2) 3
(3) 1 (4) 0

Q44. While preparing crystals of Mohr's salt, dil H_2SO_4 is added to a mixture of ferrous sulphate and ammonium sulphate, before dissolving this mixture in water, dil H_2SO_4 is added here to :

- (1) prevent the hydrolysis of ferrous sulphate (2) prevent the hydrolysis of ammonium sulphate
(3) make the medium strongly acidic (4) increase the rate of formation of crystals

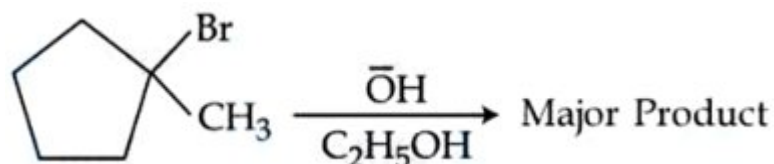
Q45. The number of complexes from the following with no electrons in the t_2 orbital is TiCl_4 , $[\text{MnO}_4]^-$, $[\text{FeO}_4]^{2-}$, $[\text{FeCl}_4]^-$, $[\text{CoCl}_4]^{2-}$

- (1) 1 (2) 4
(3) 3 (4) 2

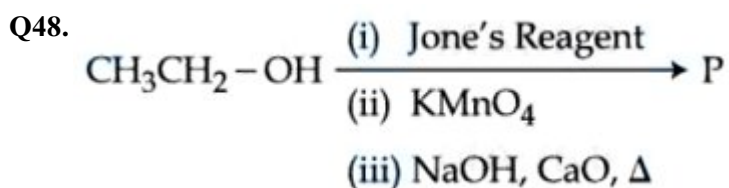
Q46. The metal atom present in the complex $MABXL$ (where A, B, X and L are unidentate ligands and M is metal) involves sp^3 hybridization. The number of geometrical isomers exhibited by the complex is:

- (1) 2 (2) 0
(3) 4 (4) 3

Q47. Identify the major product in the following reaction.



- (1) (2)
(3) (4)



Consider the above reaction sequence and identify the major product P .

- (1) Methoxymethane (2) Methanoic acid
(3) Methanal (4) Methane

Q49. Which one of the following reactions is NOT possible?

- (1)
- (2)
- (3)
- (4)

Q50. Coagulation of egg, on heating is because of :

- (1) The secondary structure of protein remains unchanged (2) Denaturation of protein occurs
- (3) Biological property of protein remains unchanged (4) Breaking of the peptide linkage in the primary structure of protein occurs

Q51. In an atom, total number of electrons having quantum numbers $n = 4$, $|m_l| = 1$ and $m_s = -\frac{1}{2}$ is _____

Q52. Number of compounds from the following with zero dipole moment is _____

HF, H₂, H₂S, CO₂, NH₃, BF₃, CH₄, CHCl₃, SiF₄, H₂O, BeF₂

Q53. Combustion of 1 mole of benzene is expressed at $\text{C}_6\text{H}_6(\text{l}) + \frac{15}{2}\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$. The standard enthalpy of combustion of 2 mol of benzene is $-x' \text{ kJ}$. $x =$ _____ Given: 1. standard Enthalpy of formation of 1 mol of C₆H₆(l), for the reaction $6\text{C}(\text{graphite}) + 3\text{H}_2(\text{g}) \rightarrow \text{C}_6\text{H}_6(\text{l})$ is 48.5 kJ mol^{-1} . 2. Standard Enthalpy of formation of 1 mol of CO₂(g), for the reaction $\text{C}(\text{graphite}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$ is $-393.5 \text{ kJ mol}^{-1}$. 3. Standard Enthalpy of formation of 1 mol of H₂O(l), for the reaction $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$ is -286 kJ mol^{-1} .

Q54. Using the given figure, the ratio of R_f values of sample A and sample C is $x \times 10^{-2}$. Value of x is _____

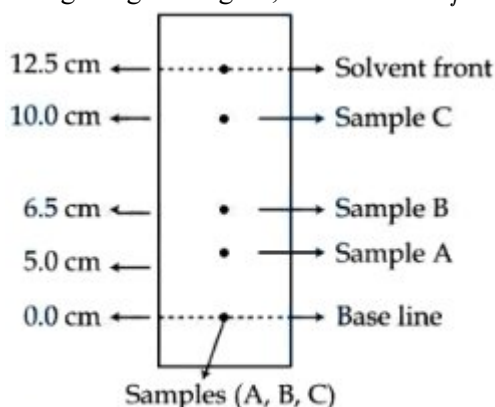


Fig : Paper chromatography of Samples

Q55. Considering acetic acid dissociates in water, its dissociation constant is 6.25×10^{-5} . If 5 mL of acetic acid is dissolved in 1 litre water, the solution will freeze at $-x \times 10^{-2}^\circ\text{C}$, provided pure water freezes at 0°C . $x =$

$$(K_f)_{\text{water}} = 1.86 \text{ K kg mol}^{-1}.$$

$$\text{density of acetic acid is } 1.2 \text{ g mol}^{-1}.$$

_____. (Nearest integer) Given : molar mass of water = 18 g mol^{-1} . Acetic acid dissociates as

$$\text{molar mass of acetic acid} = 60 \text{ g mol}^{-1}.$$

$$\text{density of water} = 1 \text{ g cm}^{-3}$$

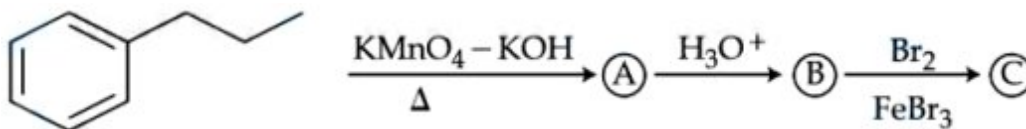


Q56. Consider the following single step reaction in gas phase at constant temperature. $2\text{A}_{(\text{g})} + \text{B}_{(\text{g})} \rightarrow \text{C}_{(\text{g})}$ The initial rate of the reaction is recorded as r_1 when the reaction starts with 1.5 atm pressure of A and 0.7 atm pressure of B. After some time, the rate r_2 is recorded when the pressure of C becomes 0.5 atm. The ratio $r_1 : r_2$ is _____ $\times 10^{-1}$. (Nearest integer)

Q57. The fusion of chromite ore with sodium carbonate in the presence of air leads to the formation of products A and B along with the evolution of CO_2 . The sum of spin-only magnetic moment values of A and B is _____ B.M. (Nearest integer) [Given atomic number : C : 6, Na : 11, O : 8, Fe : 26, Cr : 24]

Q58. In the Claisen-Schmidt reaction to prepare 351 g of dibenzalacetone using 87 g of acetone, the amount of benzaldehyde required is _____ g. (Nearest integer)

Q59. The product (C) in the following sequence of reactions has _____ π bonds.



Q60. Xg of ethanamine was subjected to reaction with NaNO_2/HCl followed by hydrolysis to liberate N_2 and HCl . The HCl generated was completely neutralised by 0.2 moles of NaOH . X is _____ g.

Q61. Let $S_1 = \{z \in \mathbb{C} : |z| \leq 5\}$, $S_2 = \left\{z \in \mathbb{C} : \text{Im} \left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i} \right) \geq 0\right\}$ and $S_3 = \{z \in \mathbb{C} : \text{Re}(z) \geq 0\}$. Then the area of the region $S_1 \cap S_2 \cap S_3$ is :

- (1) $\frac{125\pi}{12}$ (2) $\frac{125\pi}{4}$
 (3) $\frac{125\pi}{24}$ (4) $\frac{125\pi}{6}$

Q62. 60 words can be made using all the letters of the word BHBJO, with or without meaning. If these words are written as in a dictionary, then the 50th word is :

- (1) JBBOH (2) OBBJH
 (3) OBBHJ (4) HBBJO

Q63. For $x \geq 0$, the least value of K, for which $4^{1+x} + 4^{1-x}$, $\frac{K}{2}$, $16^x + 16^{-x}$ are three consecutive terms of an A.P., is equal to :

- (1) 8 (2) 4
 (3) 10 (4) 16

Q64. If the constant term in the expansion of $\left(\frac{\sqrt[5]{3}}{x} + \frac{2x}{\sqrt[3]{5}} \right)^{12}$, $x \neq 0$, is $\alpha \times 2^8 \times \sqrt[5]{3}$, then 25α is equal to :

- (1) 724 (2) 742
 (3) 639 (4) 693

Q65. Let $A(-1, 1)$ and $B(2, 3)$ be two points and P be a variable point above the line AB such that the area of $\triangle PAB$ is 10. If the locus of P is $ax + by = 15$, then $5a + 2b$ is :

- (1) 6 (2) $-\frac{6}{5}$
 (3) 4 (4) $-\frac{12}{5}$

Q66. Let ABCD and AEF G be squares of side 4 and 2 units, respectively. The point E is on the line segment AB and the point F is on the diagonal AC. Then the radius r of the circle passing through the point F and touching the line segments BC and CD satisfies:

- (1) $r = 0$ (2) $2r^2 - 4r + 1 = 0$
 (3) $2r^2 - 8r + 7 = 0$ (4) $r^2 - 8r + 8 = 0$

Q67. Let the circle $C_1 : x^2 + y^2 - 2(x + y) + 1 = 0$ and C_2 be a circle having centre at $(-1, 0)$ and radius 2. If the line of the common chord of C_1 and C_2 intersects the y -axis at the point P, then the square of the distance of P from the centre of C_1 is :

- (1) 2 (2) 1
(3) 4 (4) 6

Q68. Let the set $S = \{2, 4, 8, 16, \dots, 512\}$ be partitioned into 3 sets A, B, C with equal number of elements such that $A \cup B \cup C = S$ and $A \cap B = B \cap C = A \cap C = \phi$. The maximum number of such possible partitions of S is equal to:

- (1) 1680 (2) 1640
(3) 1520 (4) 1710

Q69. Let $\alpha\beta \neq 0$ and $A = \begin{bmatrix} \beta & \alpha & 3 \\ \alpha & \alpha & \beta \\ -\beta & \alpha & 2\alpha \end{bmatrix}$. If $B = \begin{bmatrix} 3\alpha & -9 & 3\alpha \\ -\alpha & 7 & -2\alpha \\ -2\alpha & 5 & -2\beta \end{bmatrix}$ is the matrix of cofactors of the elements of A , then $\det(AB)$ is equal to :

- (1) 64 (2) 216
(3) 343 (4) 125

Q70. The values of m, n , for which the system of equations $2x + 5y + 5z = 17$, has infinitely many solutions, satisfy the equation:

- (1) $m^2 + n^2 - mn = 39$ (2) $m^2 + n^2 - m - n = 46$
(3) $m^2 + n^2 + m + n = 64$ (4) $m^2 + n^2 + mn = 68$

Q71. Let $f, g : \mathbf{R} \rightarrow \mathbf{R}$ be defined as : $f(x) = |x - 1|$ and $g(x) = \begin{cases} e^x, & x \geq 0 \\ x + 1, & x \leq 0 \end{cases}$

Then the function $f(g(x))$ is

- (1) neither one-one nor onto. (2) one-one but not onto.
(3) onto but not one-one. (4) both one-one and onto.

Q72. Let $f : [-1, 2] \rightarrow \mathbf{R}$ be given by $f(x) = 2x^2 + x + [x^2] - [x]$, where $[t]$ denotes the greatest integer less than or equal to t . The number of points, where f is not continuous, is :

- (1) 5 (2) 6
(3) 3 (4) 4

Q73. If $y(\theta) = \frac{2 \cos \theta + \cos 2\theta}{\cos 3\theta + 4 \cos 2\theta + 5 \cos \theta + 2}$, then at $\theta = \frac{\pi}{2}$, $y'' + y' + y$ is equal to :

- (1) $\frac{1}{2}$ (2) 1
(3) 2 (4) $\frac{3}{2}$

Q74. Let $\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx$, $m, n > 0$. If $\int_0^1 (1-x^{10})^{20} dx = a \times \beta(b, c)$, then $100(a + b + c)$ equals _____

- (1) 1021 (2) 2120
(3) 2012 (4) 1120

Q75. The area enclosed between the curves $y = x|x|$ and $y = x - |x|$ is :

- (1) $\frac{4}{3}$ (2) 1
(3) $\frac{2}{3}$ (4) $\frac{8}{3}$

Q76. The differential equation of the family of circles passing through the origin and having centre at the line $y = x$ is :

- (1) $(x^2 - y^2 + 2xy)dx = (x^2 - y^2 - 2xy)dy$ (2) $(x^2 + y^2 + 2xy)dx = (x^2 + y^2 - 2xy)dy$
(3) $(x^2 + y^2 - 2xy)dx = (x^2 + y^2 + 2xy)dy$ (4) $(x^2 - y^2 + 2xy)dx = (x^2 - y^2 + 2xy)dy$

Q77. Consider three vectors $\vec{a}, \vec{b}, \vec{c}$. Let $|\vec{a}| = 2, |\vec{b}| = 3$ and $\vec{a} = \vec{b} \times \vec{c}$. If $\alpha \in [0, \frac{\pi}{3}]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c} - \vec{a}|^2$ is equal to:

- (1) 110 (2) 124
(3) 121 (4) 105

Q78. Let $\vec{a} = 2\hat{i} + 5\hat{j} - \hat{k}, \vec{b} = 2\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{c}$ be three vectors such that $(\vec{c} + \hat{i}) \times (\vec{a} + \vec{b} + \hat{i}) = \vec{a} \times (\vec{c} + \hat{i})$. If $\vec{a} \cdot \vec{c} = -29$, then $\vec{c} \cdot (-2\hat{i} + \hat{j} + \hat{k})$ is equal to:

- (1) 15 (2) 12
(3) 10 (4) 5

Q79. Let (α, β, γ) be the image of the point $(8, 5, 7)$ in the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{5}$. Then $\alpha + \beta + \gamma$ is equal to :

- (1) 16 (2) 20
(3) 14 (4) 18

Q80. The coefficients a, b, c in the quadratic equation $ax^2 + bx + c = 0$ are from the set $\{1, 2, 3, 4, 5, 6\}$. If the probability of this equation having one real root bigger than the other is p , then $216p$ equals :

- (1) 57 (2) 76
(3) 38 (4) 19

Q81. The number of real solutions of the equation $x|x+5| + 2|x+7| - 2 = 0$ is _____

Q82. If $1 + \frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}} + \frac{5-2\sqrt{6}}{18} + \frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}} + \frac{49-20\sqrt{6}}{180} + \dots$ upto $\infty = 2 + \left(\sqrt{\frac{b}{a}} + 1\right) \log_e \left(\frac{a}{b}\right)$, where a and b are integers with $\gcd(a, b) = 1$, then $11a + 18b$ is equal to _____

Q83. The number of solutions of $\sin^2 x + (2 + 2x - x^2) \sin x - 3(x-1)^2 = 0$, where $-\pi \leq x \leq \pi$, is _____

Q84. Let a line perpendicular to the line $2x - y = 10$ touch the parabola $y^2 = 4(x-9)$ at the point P . The distance of the point P from the centre of the circle $x^2 + y^2 - 14x - 8y + 56 = 0$ is _____

Q85. Let $a > 0$ be a root of the equation $2x^2 + x - 2 = 0$. If $\lim_{x \rightarrow \frac{1}{a}} \frac{16(1 - \cos(2+x-2x^2))}{(1-ax)^2} = \alpha + \beta\sqrt{17}$, where $\alpha, \beta \in \mathbb{Z}$, then $\alpha + \beta$ is equal to _____

Q86. Let the mean and the standard deviation of the probability distribution

X	α	1	0	-3
P(X)	$\frac{1}{3}$	K	$\frac{1}{6}$	$\frac{1}{4}$

be μ and σ , respectively. If $\sigma - \mu = 2$, then $\sigma + \mu$ is equal to _____

Q87. Let the maximum and minimum values of $\left(\sqrt{8x - x^2 - 12} - 4\right)^2 + (x - 7)^2, x \in \mathbf{R}$ be M and m , respectively. Then $M^2 - m^2$ is equal to _____

Q88. If $f(t) = \int_0^\pi \frac{2x \, dx}{1 - \cos^2 t \sin^2 x}, 0 < t < \pi$, then the value of $\int_0^{\frac{\pi}{2}} \frac{\pi^2 dt}{f(t)}$ equals _____

Q89. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{2x}{(1+x^2)^2} y = x e^{\frac{1}{(1+x^2)}}$; $y(0) = 0$.

Then the area enclosed by the curve $f(x) = y(x) e^{-\frac{1}{(1+x^2)}}$ and the line $y - x = 4$ is _____

Q90. Let the point $(-1, \alpha, \beta)$ lie on the line of the shortest distance between the lines $\frac{x+2}{-3} = \frac{y-2}{4} = \frac{z-5}{2}$ and $\frac{x+2}{-1} = \frac{y+6}{2} = \frac{z-1}{0}$. Then $(\alpha - \beta)^2$ is equal to _____

ANSWER KEYS

1. (3)	2. (3)	3. (4)	4. (2)	5. (2)	6. (2)	7. (2)	8. (3)
9. (3)	10. (4)	11. (3)	12. (2)	13. (1)	14. (3)	15. (1)	16. (3)
17. (2)	18. (2)	19. (3)	20. (2)	21. (16)	22. (2)	23. (1000)	24. (60)
25. (16)	26. (5)	27. (500)	28. (4)	29. (275)	30. (6588)	31. (4)	32. (4)
33. (1)	34. (1)	35. (2)	36. (3)	37. (2)	38. (4)	39. (3)	40. (3)
41. (2)	42. (4)	43. (2)	44. (1)	45. (3)	46. (2)	47. (3)	48. (4)
49. (4)	50. (2)	51. (6)	52. (6)	53. (6535)	54. (50)	55. (19)	56. (315)
57. (6)	58. (318)	59. (4)	60. (9)	61. (1)	62. (2)	63. (3)	64. (4)
65. (4)	66. (4)	67. (1)	68. (1)	69. (2)	70. (1)	71. (1)	72. (4)
73. (3)	74. (2)	75. (1)	76. (1)	77. (2)	78. (4)	79. (3)	80. (3)
81. (3)	82. (76)	83. (2)	84. (10)	85. (170)	86. (5)	87. (1600)	88. (1)
89. (18)	90. (25)						